Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29 October 2026
Team ID	N/A - individual project; no Team ID provided
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Data preparation, web routes and database queries.	Python, SQL, HTML, CSS, JavaScript	Python/pandas for preparation and Flask; SQL for SQLite aggregation; HTML/CSS/JS for website and embeds.
2	Database / Storage	Structured store for cleaned project records.	SQLite (ev_market.db)	SQLite supports reproducible local analytical queries; the table ev_market_clean contains 199,999 rows.
3	Data Ingestion	Load cleaned records from the project data file.	CSV import / Python preparation script	The cleaned CSV is the Tableau Public data connection; SQLite DB is loaded for SQL/API tasks.
4	Transformation & Business Logic	Deduplication, median imputation, type/quality checks, grouping and calculated metrics.	Python/pandas, SQLite SQL, Tableau calculated fields	Documented preprocessing and explicit aggregations support comparisons.
5	Data Visualization / Analytics	Nine chart worksheets, dashboard and three-scene Story.	Tableau Public Desktop / Tableau Public	Interactive filters and view-level tooltips make grouped patterns explorable.
6	Front-End / User Interface	Public website hosts dashboard and Story.	Flask, HTML/CSS/JavaScript, Tableau Embedding API v3	The site provides two embedded view areas and supporting API routes; deployed on Render.